

Guide: Deploy to AWS Elastic Beanstalk

Step 1: Create a Deployment Package

To deploy the app, AWS Elastic Beanstalk needs to receive a ZIP of your source code.

1. Zip everything inside your project directory (do NOT zip the project itself).
2. Make sure the ZIP file is compressed (e.g. hex-calculator.zip)

Step 2: Launch Elastic Beanstalk

1. Login to your **AWS Management Console**.
2. Search for and select **Elastic Beanstalk**.
3. Click the **Create application** button.

Step 3: Configure Your Application

Complete the configuration wizard with these values:

Application Name: hexadecimal calculator

Platform: Docker

Platform Branch: Docker running on 64bit Amazon Linux

Application Code: Choose **Upload your code**, choose **Local file** and use the ZIP file you compressed in Step 1.

Step 4: Configure Service Access

If AWS asks you for roles and permissions, the default permissions are:

Service Role: Choose aws-elasticbeanstalk-service-role (or let AWS create one)

EC2 Instance Profile: Choose aws-elasticbeanstalk-ec2-role (or let AWS create one)

Step 5: Review and Create

1. Click **Next** for the next screens. The defaults are fine for this project:
 - The default networking is fine.
 - The default capacity (one instance) is fine.
 - The default monitoring is fine.
2. Review your selections and click **Submit** or **Create Environment**.

Step 6: Wait for the Deployment

AWS now automatically configures the infrastructure for you. This will take a couple of minutes while elastic beanstalk starts the following pipeline in the background:

When the deployment process completes, Elastic Beanstalk will show you a public URL at the top of the dashboard:

Click on that link to launch your live, cloud-based Hex Calculator in your browser!